The Hidden Cost of Discounts

Eric Chen, Lawrence Du, Peyton Garrett, Sam Reade, Su-Han Tan

March 2026

1 Introduction

With growing demands and competition in the retail market, businesses are constantly seeking optimal discounting strategies to maximize profit. While discounts may stimulate consumer demand, they also reduce margins, creating uncertainty about whether the strategy actually improves profitability. This project investigates whether company discount strategies are profit-optimizing and whether it differs across US states utilizing retail transaction data from the Superstore dataset.

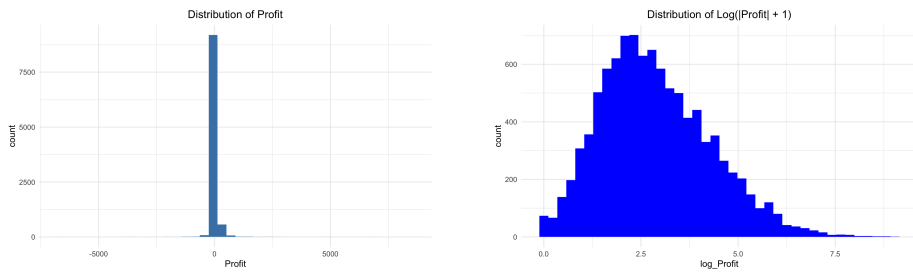
To address this question, we applied various statistical techniques. We did initial exploratory data analysis to understand the distribution of key variables like profit, sales, and discount levels. We then tested whether orders with discounts differ significantly in profitability compared to non-discounted items using statistical inference. Leveraging various statistical models, we analyze how profit varies with discount levels and US state controlling for various variables. Lastly, we identified product sub categories where current discounting strategies may be harmful to profitability by estimating the ideal break-even thresholds, and simulating various discount caps to understand effects on overall profitability.

2 Data Wrangling

Before our analysis, data wrangling steps were performed to prepare the dataset for our statistical investigation. To better capture profitability, we created a profit margin variable defined as profit divided by sales and a binary indicator identifying transactions that resulted in a loss. Identifier variables such as Row.ID, Customer.ID, and Product.ID were removed because they serve only as labels. Finally, categorical variables including Segment, Region, Category, Sub.Category, and Ship.Mode were converted to factor types to support grouped analysis and statistical modeling. These steps ensured the dataset was clean and suitable for further analysis.

3 Exploratory Data Analysis

Exploratory Data Analysis was conducted to examine the distributions and relationships among important variables in the Superstore dataset. Descriptive statistics and visualizations showed that Profit is highly right-skewed. This skewness was confirmed by a skewness statistics of approximately 7.56 for Profit, indicating a large deviation from normality. A Shapiro–Wilk normality test was also applied to check for normality of the Profit variable. The test rejected the assumption of normality with an extremely low p-value. To mitigate the influence of extreme values and improve the variable distribution, a logarithmic transformation was applied to Profit. After the transformation the skewness statistics were improved to 0.54 for Profit.



(a) Distribution of Profit

(b) Distribution of $\log(|\text{Profit}| + 1)$

Figure 1: EDA Distributions

Additionally, a correlation matrix was used to explore relationships between numeric variables including Sales, Profit, Discount, and Quantity. The results indicate a moderate positive relationship between Sales and Profit, with a positive correlation of approximately 0.48, which suggests that larger sales tend to generate higher profits overall. However, Discount shows a negative correlation of 0.22 with Profit, indicating that higher discounts are generally associated with lower profits. This pattern suggests that aggressive discounting may reduce profit margins.

Figure 2: Correlation Matrix

4 Statistical Inference

To formally evaluate whether discounted orders differ significantly in profitability compared to non-discounted orders, we conducted a series of statistical inference analyses. These analyses combine descriptive statistics, visualization, and hypothesis testing to determine whether the observed differences in profitability are statistically significant.

We first separated the observations into two groups based on discount status. Orders with a positive discount were classified as discounted, while orders with zero discount were classified as non-discounted. We then compared the profitability of these two groups using profit and profit margin as outcome variables.

Before performing hypothesis testing, we computed summary statistics for both groups. The results indicate a substantial difference in profitability. The average profit for non-discounted orders is 66.90, whereas the average profit for discounted orders is -6.66. Similarly, the mean profit margin for non-discounted orders is 0.340, compared to -0.083 for discounted orders. These statistics suggest that discounted orders tend to generate substantially lower profitability.

Discounted	n	mean_profit	sd_profit	median_profit
No	4798	66.900293	257.0554	15.9952
Yes	5196	-6.657156	204.8135	3.3408

Discounted	n	mean_margin	sd_margin	median_margin
No	4798	0.3401605	0.1346967	0.34
Yes	5196	-0.0826930	0.5625465	0.10

Figure 3: Two Sample Inference Comparison Tables

To visualize these differences, we constructed boxplots comparing profit and profit margin distributions between discounted and non-discounted orders. The boxplots reveal that discounted transactions exhibit a much wider spread and a larger number of negative profit observations. In contrast, non-discounted orders tend to have more consistently positive profits and margins.

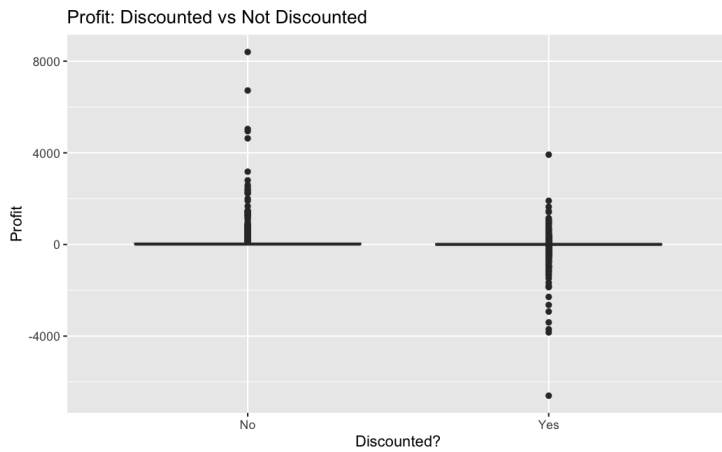


Figure 4: Profit Boxplot

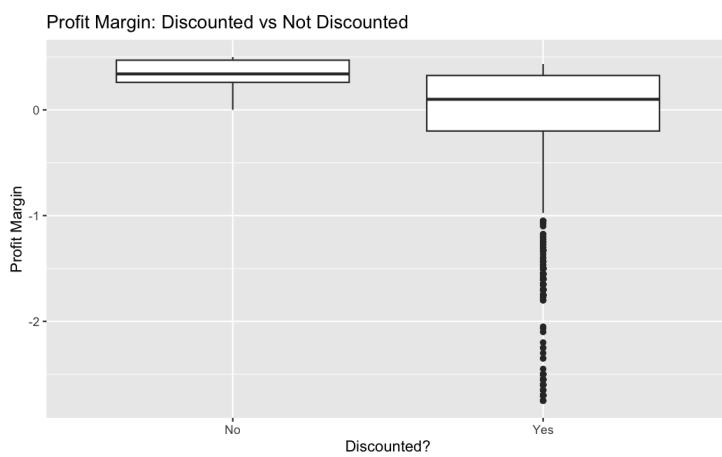


Figure 5: Profit Margin Boxplot

For profit, the Welch t-test produces a test statistic of $t = 15.738$ with $p < 0.001$, indicating a highly significant difference in mean profit between the two groups. The 95% confidence interval for the difference in means ranges from 64.40 to 82.72, suggesting that non-discounted orders generate substantially higher profit on average.

Because retail transaction data may exhibit skewness and outliers, we also conducted a Wilcoxon rank-sum test as a non-parametric robustness check. The Wilcoxon test also yields $p < 0.001$, confirming that the profit distributions differ significantly between discounted and non-discounted orders.

We repeated the same inference procedure using profit margin as the outcome variable. The Welch t-test for profit margin produces $t = 52.576$ with

$p < 0.001$, and the estimated difference in mean margins lies between 0.407 and 0.439. The Wilcoxon test again confirms this result with $p < 0.001$, indicating a consistent difference in profitability across both parametric and non-parametric inference methods.

Test	Statistic	DF	P_Value
Welch Two Sample t-test (Profit)	15.73799	9162.244	0

Test	Statistic	DF	P_Value
Welch Two Sample t-test (Profit Margin)	52.57579	5835.756	0

Figure 6: Two Sample t-test Results

Taken together, these results provide strong statistical evidence that discounted orders are associated with significantly lower profitability. Both profit levels and profit margins are substantially lower for discounted transactions, and the conclusions remain robust across different statistical tests.

However, it is important to interpret these results carefully. The statistical inference analysis demonstrates a strong association between discounting and lower profitability, but it does not establish a causal relationship. Discounted orders may differ systematically from non-discounted orders in other characteristics, such as product category, region, or order size. For this reason, the following section introduces regression models and ANOVA analyses to examine whether these relationships remain after controlling for additional variables.

5 Statistical Modeling

To evaluate whether the impact of discounts differ across states in the US, we conducted a two-way analysis of variance (ANOVA) using profit as the outcome variable. The model includes whether an order had a discount and state as explanatory variables, along with an interaction term between them.

$$Total_Profit_{ij} = \beta_0 + \beta_1 Discount_i + \beta_2 State_j + \beta_3 (Discount_i \times State_j) + \epsilon_{ij}$$

Where:

- $Total_Profit_{ij}$ = total profit for order i in state j
- $Discount_i$ = binary indicator (1 = discount, 0 = no discount)
- $State_j$ = state where the order occurred
- ϵ_{ij} = error term

In this framework, the coefficient for Discount measures the difference in average profit in between discounted and non-discounted orders, the state variable captures differences in profitability across geographic locations, and the interaction term tests whether the effect of offering a discount differs across states. The two-way ANOVA allows us to compare mean profits across groups defined by both discount status and state.

term	df	sumsq	meansq	statistic	p.value
Discount_Binary	1	15530440	15530439.7	140.468	0.000
State	48	32152144	669836.3	6.058	0.000
Discount_Binary:State	16	1671529	104470.6	0.945	0.516
Residuals	4943	546508007	110562.0	NA	NA

Figure 7: Two-Way ANOVA Results

The results of the ANOVA show that both discount status and state are statistically significant predictors of profit. This indicates that the presence of a discount is associated with a change of average profit and that average profit levels differ across states. However, the interaction between discount and state is not statistically significant. This suggests that the relationship between discounts and profit is **consistent across states**. In other words, while discounts influence profitability and some states are more profitable than others, **the effect of offering a discount does not differ from state to state**.

To evaluate whether the impact of discounts differ across **regions** in the US, we conducted a multiple linear regression using profit as the outcome variable. We created four models that all include whether an order had a discount and region as explanatory along with an interaction term between discount and region. In order to determine if these variables were truly significant we created combinations of the model with quantity and sub category as variables to see if those variables had any effects on the significance of the variables we were interested in.

Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$	
(Intercept)	91.939	8.018	11.467	$< 2 \times 10^{-16}$	***
Discount	-116.300	9.995	-11.636	$< 2 \times 10^{-16}$	***
RegionEast	-19.215	10.051	-1.912	0.0559	.
RegionSouth	-13.701	11.420	-1.200	0.2303	
RegionWest	-47.357	9.763	-4.851	1.25×10^{-6}	***
Discount:RegionEast	33.672	13.217	2.548	0.0109	*
Discount:RegionSouth	18.145	15.210	1.193	0.2329	
Discount:RegionWest	93.180	12.912	7.217	5.72×10^{-13}	***

Figure 8: Linear Regression Results: Effects of Discount and Region

The results of all the multiple linear regression models show that both dis-

count status and region, particularly the west region, are statistically significant predictors of profit. This indicates that the presence of a discount is associated with a change of average profit and that average profit levels differ across the North and West regions. While state level variation did not significantly alter the effect of discounts, the interaction between discount and region is statistically significant. This suggests that orders are affecting how discounts affect profit in comparison to the orders differently across the North and West regions. North and West regions differ in profitability and the effect of offering a discount does differ across the North and West regions. Since the discounts and regionwest are statistically significant, we can conclude that orders from the west region decrease profit in comparison to the north region as well as discounts.

6 Business Implications

To translate our statistical findings into actionable business insights, we evaluate whether the current discount practices are profit-optimizing across the various sub categories. Our analysis identifies categories where high discounts are harmful for the business, so we estimated the breakeven thresholds to determine the point at which discounts turn profit negative or positive for the 17 different sub categories. By creating a model that predicts the profit from discount, we were able to estimate the breakeven point for every sub-category by taking the negated value of the model intercept divided by the model slope. After, we calculated the discount gap by subtracting the breakeven discount rate from the current average discount rate for that sub-category.

Sub-Category	Avg Discount	Break-even Discount	Discount Gap
Fasteners	0.082	0.581	-0.499
Envelopes	0.080	0.494	-0.414
Paper	0.075	0.471	-0.396
Binders	0.372	0.457	-0.085
Labels	0.069	0.389	-0.320
Copiers	0.162	0.367	-0.205
Machines	0.306	0.318	-0.012
Appliances	0.167	0.311	-0.145
Art	0.075	0.311	-0.236
Accessories	0.078	0.282	-0.204
Phones	0.155	0.277	-0.122
Furnishings	0.138	0.252	-0.114
Chairs	0.170	0.232	-0.062
Tables	0.261	0.200	0.061
Bookcases	0.211	0.185	0.026
Storage	0.075	0.156	-0.082
Supplies	0.077	0.054	0.023

Figure 9: Subcategory Discount Table

Tables, Bookcases, and Supplies currently exhibit discount rates higher than their ideal breakeven point, which may indicate excessive discounting. However, for the other 14 categories, they are generally discounted below the breakeven point, so they can sustain moderate discounts. This table can guide the business in understanding where to focus its discounting efforts and resources by understanding an estimated break even discount rate for every sub-category.

Furthermore, we wanted to understand the impact of hypothetical discounting policy changes, such as setting general discount caps pertinent across all categories. We simulated 21 different discount caps between 10% and 30% to understand how the caps would affect overall profit. The results suggested that a 20% cap seems to be optimal for thin-margin products, as we can see from the graph below that the simulated total profit starts to dip very drastically after a 20% discount. Caps above 20% may excessively erode profit, while lower caps may de-incentivize customers to continue purchasing from the business. Note that this is a general cap, but the business should also use the aforementioned table to inform business decisions.

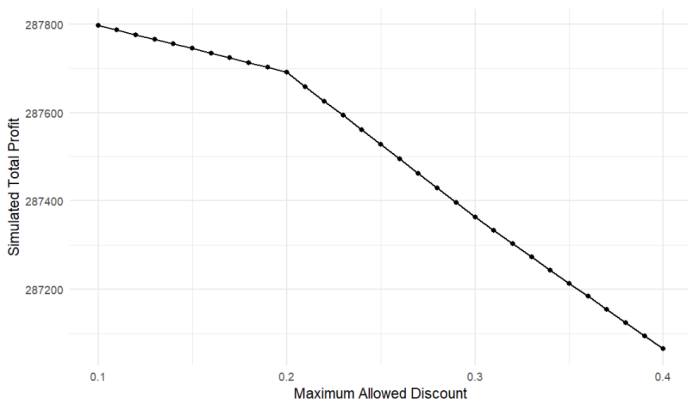


Figure 10: Simulated Total Profit with 0.1 to 0.4 Discount Rate (10%-40%)

In this graph below, we can also understand the relationship between quantity sold and the discount rate. At 20% discount, the total quantity suddenly spikes back up to the maximum point besides a 0% discount, which may be indicative of consumer behavior and further corroborates our previous notion that a 20% discount cap is reasonable.

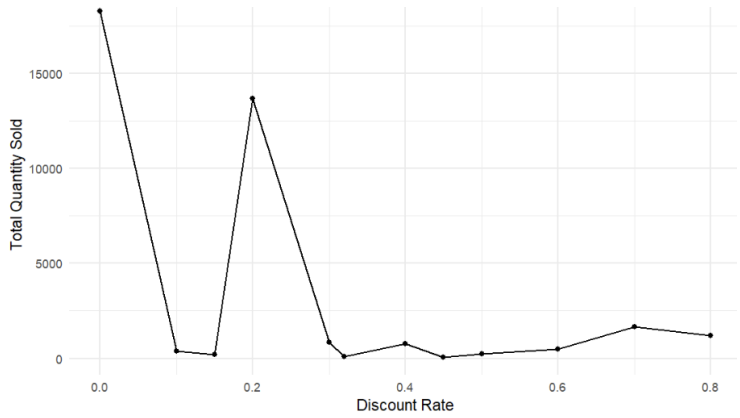


Figure 11: Total Quantity of Items Sold at the Current Discount Rates

7 Conclusion

This study examined whether current discounting strategies are profit optimizing in the Superstore dataset. We found strong evidence that discounted orders are associated with significantly lower profits and profit margins compared to non-discounted orders. The effect of discounting on profit is mostly consistent across the 49 states in our datasets, with differences on the regional level. Several subcategories like tables, bookcases, and supplies are currently discounted above our projected breakeven thresholds, suggesting there are opportunities to discount less aggressively on these items to still entice more customers while remaining profitable.

Our simulation also demonstrates a discount cap around 20% may be an optimal point to maintain customer loyalty without excessively eroding margins. However, this recommendation is an overarching recommendation across all 17 subcategories, so it should be taken cautiously as there are some items or even seasons of the year where this policy may be harmful. Overall, our findings suggest that retailers should adopt more targeted discounting strategies, focusing on categories that can sustain moderate discounts while minimizing aggressive discounts on the thinner-margin products. Continuous monitoring of individual sub-category level profitability will allow the business to adjust discounts more effectively, ensuring maximization of profit without eroding customer loyalty.

8 References

Chowdhury, V. (2022, February 17). Superstore dataset. Kaggle.
https://www.kaggle.com/datasets/vivek468/superstore-dataset-final